To such λ we will attach a **Young diagram** Y_λ, which is the union of rectangles $-j \le y \le -j+1$, $0 \le x \le \lambda_j$ in the coordinate plane, for $j = 1, \dots, p$. Clearly, Y_λ is the union of n unit squares $(i,j) := \{(x,y) \in \mathbb{R}^2 | -j \le x \le -j+1, i-1 \le x \le i\}$, $j = 1, \dots, p$, $i = 1, \dots, \lambda_j$. A **Young tableau** corresponding to Y_λ is the result of filling the numbers $1, \dots, n$ into the squares of Y_λ in some way (without repetitions). For example, we will consider the Young tableau T_λ obtained by filling in the numbers in increasing order, left to right, top to bottom.

We can define two subgroups of S_n corresponding to T_λ:

1. The row subgroup P_λ: the subgroup which maps every element of $\{1, \dots, n\}$ into an element standing in the same row in T_λ.

2. The column subgroup Q_λ: the subgroup which maps every element of $\{1, \dots, n\}$ into an element standing in the same column in T_λ.

Clearly, $P_\lambda \cap Q_\lambda = \{1\}$.

Define the **Young projectors**

$$a_\lambda := \frac{1}{|P_\lambda|} \sum_{g \in P_\lambda} g,$$

$$b_\lambda := \frac{1}{|Q_\lambda|} \sum_{g \in Q_\lambda} (-1)^g g,$$

where $(-1)^g$ denotes the sign of the permutation g. Set $c_\lambda = a_\lambda b_\lambda$. Since $P_\lambda \cap Q_\lambda = \{1\}$, this element is nonzero.

The irreducible representations of S_n are described by the following theorem.

Theorem 5.12.2. *The subspace $V_\lambda := \mathbb{C}[S_n]c_\lambda$ of $\mathbb{C}[S_n]$ is an irreducible representation of S_n under left multiplication. Every irreducible representation of S_n is isomorphic to V_λ for a unique λ.*

The modules V_λ are called the **Specht modules**.

The proof of this theorem is given in the next subsection.

Example 5.12.3. For the partition $\lambda = (n)$, $P_\lambda = S_n$, $Q_\lambda = \{1\}$, so c_λ is the symmetrizer, and hence V_λ is the trivial representation.